

PIBC: Think like a housing economist

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The time value of money

- A dollar now generally viewed as more valuable than a dollar in the future
 - Inflation
 - Can invest the dollar now and earn a return
 - Impatience
- A discount rate translates future to present dollars
 - We will call this rate d
 - The “present value” of a payment of $\$x$ in t years is $\frac{x}{[1+d]^t}$.
 - Example if discount rate is 10% \$1,000 in 7 years has PV

$$\frac{1000}{[1.10]^7} \approx \$513$$

- Project Net Present Value (“NPV”)
 - Is sum of PV associated with project at all dates
 - Including negative of investment today (date 0)

$$NPV = \sum_{t=0}^{\infty} x_t [1 + d]^{-t}$$

More on discounting

- Arguably, parents job to inculcate low discount rate
- IRR: “internal rate of return”
 - Like NPV Jeopardy
 - “At this discount rate, NPV of project is zero”
 - Positive IRR only if $IRR > d$
- Project pro formas
 - All the info needed to compute NPV and IRR
 - Much more detail than just dates and cash flows
 - Important ratio: “cap rate”
 - Net operating income (NOI) in year 1 divided by cost
Acquisition cost for simple purchase
Total project cost o/w
 - NOI vs cashflow
 - NOI ignores financing (interest and amortization)
 - Cashflow $\approx$ NOI after financing costs

The world's simplest pro forma

- Buy land for \$1M today
- Spend \$1M building something also today
- Sell for \$2.25M in one year
- Discount rate: 10%
- NPV: $-1M - 1M + \frac{2.5M}{1.1} \approx \$272K$
- IRR: $-2M + 2.5M/(1+IRR) = 0 \Rightarrow IRR=25\%$

Useful game to understand discounting

- Pick your favourite musician's last name
 - A-M: $d = 3\%$
 - N-Z: $d = 10\%$
- Match up with a partner
- Try to create a loan that has positive NPV to both of you
- Loan has amount L , interest rate r , and term of 1 year
- Borrower NPV: $L \left[1 - \frac{1+r}{1+d} \right]$
 - Positive iff $r < d$
- Lender NPV: $L \left[\frac{1+r}{1+d} - 1 \right]$
 - Positive iff $r > d$
- If same discount rate, impossible to have +NPV for both

Segue: from game to why affordable housing mandates are dicey

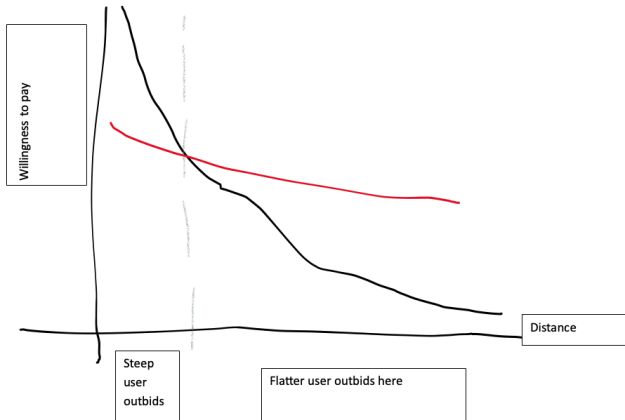
- What do markets do, ideally?
- Get everyone to agree on ratios of benefits to cost
- I have a discount rate of 10%, you 5%
 - You should lend me money at a rate between 5% and 10%
 - Positive NPV to both of us
- Very similar:
 - You can build an apartment for \$500K
 - I value the apartment at \$550K
- With “complete markets” trades happen until agreement
 - I have so much housing, I no longer value more than your cost
- Affordability mandates:
 - Someone willing to pay \$400K
 - Market price \$600K
 - Force sale at, say, \$350K
 - Buyer gets +\$50K, seller gets -\$250K
 - “Pareto gain”: seller gives buyer \$150K!

The money value of time and urban land markets

- “Location, location, location”
- The classical “monocentric city model”
 - Everyone commutes to location 0: the CBD
 - Cost of commuting: t per unit of distance
 - How does land price change with distance?
- “Bid-rent curves”
 - Willingness to pay for land at distance x is $R(x)$
 - To be indifferent between living at x a tiny bit further
 - Consume $I(x)$ units of land
 - $R(x)I(x) = R(x + dx)I(x + dx) + t \cdot dx$
 - So slope of R curve is $-\frac{t}{I(x)}$
 - Land valuation falls with distance,
 - But more slowly if you consume more land
 - Hence steeper curve closer to downtown
 - (also t bigger downtown)

Equilibrium with multiple uses

- Remember, slope of bid-rent curve is $-\frac{t}{I(x)}$
- Closer to downtown will be steeper curve



Who should be closer to town given bid rent curve logic?

- Remember, slope of bid-rent curve is $-\frac{t}{I(x)}$
- Starbucks vs WalMart
- Yuppie versus family with 1 earner, 2 kids, 2 pets?
- Apartment building versus house with yard?
- Office building versus warehouse?

Cap rates, elasticity of supply, and equilibrium
Where should cap rate be lower?
Inelastically supplied city if demand is growing

